

Response to the Third-Round Review

JBCB-1505, third major revision

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Manuscript title: Locating resolution-dependent behavioral correspondence in qualitative DNA-damage response models.

We thank the reviewer for accepting the mathematical development, which remains unchanged. This revision focuses on the biological demonstration: the death-receptor comparison is now the principal case and GIM a secondary illustration. The answers below state the models, results and limitations without requiring local documentation.

1. The GIM result is largely predetermined by the labels

We agree with the substance of this concern. Interface C exposes mechanistic differences already specified in the models; the method does not discover those terminal mechanisms. Section 5 is now entitled “GIM illustration: interface-conditioned behavioral correspondence.” It reports weak correspondence at A/B and its first loss at C among the three tested interfaces, with unchanged PN-GDDA similarity 0.9976. The abstract, Introduction, Discussion and Conclusion no longer present GIM as the principal scientific demonstration or as locating a unique biological resolution. The full Petri-net diagrams and supporting evidence remain in the article.

2. Models should agree on ordinary predictions but differ in future choices

We added a fixed-interface comparison derived from the public original model of Calzone et al., *Mathematical Modelling of Cell-Fate Decision in Response to Death Receptor Engagement*, PLoS Computational Biology 6:e1000702 (2010), DOI: 10.1371/journal.pcbi.1000702. Its original publication and Figures S2/S3 explicitly study removal of CASP3-to-CASP8 feedback and the effect of transient versus sustained ligand. DR-FB+ retains the published rules. DR-FB- removes only the CASP3 contribution to CASP8 activation; CASP3 itself remains dynamic. The shared initial condition has TNF=FADD=ATP=cIAP=1 and all other nodes zero. Updates are unitary asynchronous. NFkB, CASP3 and MPT activation/deactivation are the fixed observable events, and withdrawal changes only TNF from one to zero with identical availability in both models. No interface was retuned to obtain a desired relation.

Ordinary baseline. Complete reachable-state enumeration gives exactly the same three stable-fate predictions under sustained TNF: survival, apoptosis and necrosis. However, global sustained trace equality is now refuted, not merely unresolved. An expanded search found the 12-action word

$$\begin{aligned} &\text{MPT}\uparrow, \text{CASP3}\uparrow, \text{CASP3}\downarrow, \text{NFkB}\uparrow, \\ &\text{NFkB}\downarrow, \text{MPT}\downarrow, (\text{CASP3}\uparrow, \text{CASP3}\downarrow)^3, \end{aligned}$$

where the superscript repeats the pair three times. It occurs in DR-FB+ but not DR-FB-. Independent sparse reachability and mCRL2 confirm this decision; a 53-update accepting path was also checked against the full 28-node rules. Consequently DR-FB- cannot weakly simulate DR-FB+,

and the global baseline systems are not weakly bisimilar. The reverse trace inclusion remains undecided. These findings are stated in the abstract, Figure 1, main Results and Discussion. The word is a model prediction, not experimental evidence.

Specific shared history. The automatic search identifies a jointly reachable retained state after eight updates: TNFR, DISC_TNF, CASP8, BAX, MOMP, Cyt_c, apoptosome and CASP3 activation. Only the last update is visible at the declared interface, so the common observable history is $h = \text{CASP3_up}$. At this state, TNF, FADD, ATP, cIAP and all eight listed components are one; the other retained components are zero. The three omitted nodes, NonACD, Apoptosis and Survival, are unobserved output-only copies with no feedback. Their deletion preserves weak behavior, and the full 28-node conditioned systems were also checked directly.

Exact conditioned baseline and intervention result. From the specified shared state, the two sustained continuations each have three states and two silent edges. Their exact observable language is $\{\epsilon\}$, and they are strongly and weakly bisimilar. Once withdrawal is permitted, the continuations have 12/17 and 34/54 states/edges. Writing them K_+ and K_- , with $X \preceq_w Y$ meaning that Y simulates X , we find

$$K_+ \preceq_w K_-, \quad K_- \not\preceq_w K_+, \quad K_+ \not\preceq_w K_-.$$

The word `withdraw_TNF`, `CASP3_down` occurs only in K_- . The independent game certificate eliminates the initial pair only after covering all possible defending weak replies. It does not infer a failed global relation from one arbitrarily chosen successor pair. Production predicates, the independent game and mCRL2 equivalence checks agree.

After immediate withdrawal from this shared state, K_+ retains CASP3 on every continuation and has only an apoptotic terminal fate. K_- has only the naive terminal signature. Both continuation graphs are acyclic; their post-withdrawal outcome does not require an added fairness assumption. “Naive” is a logical marker signature, not evidence that a cell can recover after executing apoptosis.

Visible history versus hidden state. The history h is compatible with 78 internal states in each model, not just the selected state. Complete aggregation after immediate withdrawal yields

$$\begin{aligned} \text{Futures}_+(h) &= \{\text{apoptosis, naive, necrosis}\}, \\ \text{Futures}_-(h) &= \{\text{naive, necrosis}\}. \end{aligned}$$

Thus a future-set difference also exists at the shared-history level, but CASP3 activation alone does not establish unique commitment in every compatible feedback-positive state. Section 4 distinguishes this aggregate claim from the fully specified state’s persistent-CASP3 certificate.

What is not established. The complete intervention-aware languages are unequal: exact counterexamples refute both global trace inclusions and, consequently, both global weak simulations. For example, `CASP3_up`, `withdraw_TNF`, `CASP3_down` is possible only in DR-FB-. In the other direction, `withdraw_TNF`, `CASP3_up`, `CASP3_down`, `CASP3_up` is possible only in DR-FB+. Trace comparison can therefore distinguish the models both before and after withdrawal is included. We do not present this case as proving that branching alone distinguishes globally trace-equivalent biological models. It supplies the weaker, biologically grounded result of matching baseline endpoints and exact conditioned sustained continuations, followed by different intervention futures. The stronger biological counterpart of the accepted early/late-choice example remains an open limitation.

3. Identify a meaningful perturbation or observation

The new Figure 1 and Section 4 identify the logic of a pulse/withdrawal comparison at a CASP3-positive stage. Subsequent joint readout of CASP3, NFkB and MPT can distinguish persistent apoptotic activity from deactivation or movement toward the necrotic branch; ATP supports terminal classification. Monitoring all three matters because a path with MPT activation before CASP3 loss is not the same as isolated CASP3 deactivation after withdrawal. The history alone does not identify the internal witness, so the state-specific prediction would require additional state assessment. We provide neither unsupported physical timings nor an invented selective inhibitor protocol.

The source publication already studied ligand withdrawal and the feedback mechanism. Our contribution is the reproducible shared-state/history construction, future-set comparison and checked relational explanation. This is a recovery and characterization of a published phenomenon, not a new wet-lab discovery. No new experimental confirmation is claimed.

4. Qualify the claimed biological boundary

The GIM interpretation is now centrally restricted to the first loss of correspondence among Interfaces A, B and C. It is not a unique, globally minimal or experimentally established biological boundary. No exhaustive search over observational interfaces was performed. Section 8 separates this interface-change question from the fixed-interface withdrawal question.

5. Distinguish specified knowledge from computed findings

The main article now includes an explicit evidence-status table: the feedback mechanism is a published assumption; the marker interface and withdrawal semantics are declared inputs; stable-fate sets, relation decisions and witness histories are computed outputs; the discriminating observation is model-derived; wet-lab confirmation is absent. The local prospective configuration is not presented as externally registered. Resource-driven changes remove only provably unobserved output sinks and use compact storage; they do not change observables or biological rules.

6. Preserve the accepted mathematics and ensure reproducibility

All accepted definitions, propositions, proofs, the fixed-point algorithm and the early/late-choice example are retained, with an automated source comparison checking their preservation. The new witness engine is tested on the accepted example, including rejection of a certificate that omits a defending reply. The new biological computation independently verifies compact execution, terminal reachability, marker persistence and both simulation directions. The eight-update commitment state is found by an exhaustive depth scan; at that depth, one of 73 states has persistent CASP3 and a unique apoptotic terminal fate after withdrawal. Counts are not biological probabilities, and update depth is not physical time.

Section 7 and Supplementary Validation report the computational checks, source provenance, semantics, limits and certificates.